

# Public Dataset Discovery — MELAS Syndrome (MONDO:0010789)

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## Summary

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MELAS (Mitochondrial Encephalomyopathy, Lactic Acidosis, and Stroke-like episodes) is a maternally inherited mitochondrial disease, most often caused by the heteroplasmic **m.3243A>G** variant in **MT-TL1** (mt-tRNA-Leu(UUR)), with rarer variants (e.g. m.14453G>A in MT-ND5, m.3243A>T, MT-CO3). Public omics data for MELAS **does exist** and is reasonably rich for a rare Mendelian disorder, but it is **dominated by patient-derived cellular models of m.3243A>G** (fibroblasts, iPSC-derived neurons/endothelium/myofibres, organoids, cybrid-like models). Primary human patient tissue is limited to a few blood, skeletal-muscle biopsy, and post-mortem eye datasets. Below are datasets verified by direct query against repository APIs (NCBI GEO/E-utilities & BioProject, EBI BioStudies/PRIDE, EBI Search for MetaboLights, and MassIVE). Every accession below was resolved to a live record.

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## Datasets — primary patient material

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### GSE1462

- **Repository:** GEO (also GDS1065; BioProject PRJNA90037)
- **Title:** Mitochondrial disorders
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 15 (includes 4 A3243G-MELAS subjects, plus A3243G-PEO, common-deletion, and normal)
- **Sample types:** Skeletal muscle biopsy
- **Conditions:** A3243G-MELAS vs A3243G-PEO vs mtDNA common-deletion vs normal

- **Publication:**

**P** 15728662

- **Mechanistic relevance:** One of the earliest transcriptomic comparisons of the same m.3243A>G variant across two clinical presentations (MELAS vs PEO) in patient muscle, directly addressing genotype–phenotype divergence in mitochondrial encephalomyopathy. It profiles the OXPHOS/energy-metabolism transcriptional response in the affected tissue (muscle) central to lactic acidosis and exercise intolerance.
- **Confidence in accession:** HIGH

## GSE14882

- **Repository:** GEO (ArrayExpress mirror E-GEOD-14882; BioProject PRJNA111945)
- **Title:** Expression data from human blood from MELAS patients and controls
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 16
- **Sample types:** Whole blood (peripheral)
- **Conditions:** MELAS patients (A3243G) vs controls
- **Publication:**

**P** 21708074

- **Mechanistic relevance:** A rare example of primary patient (blood) transcriptomics; the study links the A3243G heteroplasmic burden to reconfiguration of blood gene-expression programs, informing systemic/peripheral signatures and potential biomarkers of MELAS.
- **Confidence in accession:** HIGH

## GSE42986

- **Repository:** GEO
- **Title:** Transcriptome profiling in human primary mitochondrial respiratory chain disease
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 53 (mitochondrial respiratory-chain disease cohort containing MELAS m.3243A>G fibroblast strata, e.g. "Definite Mito Disease — MELAS - m.3243A>G")

- **Sample types:** Primary dermal fibroblasts (and flash-frozen skeletal muscle)
- **Conditions:** Multiple respiratory-chain disease genotypes (incl. MELAS) vs controls
- **Publication:**

**P** 23894440

- **Mechanistic relevance:** Identifies a common transcriptional response to respiratory-chain dysfunction; the MELAS m.3243A>G subset lets one isolate the shared bioenergetic-stress signature from variant-specific effects, relevant to the OXPHOS failure driving MELAS.
- **Confidence in accession:** HIGH

## GSE202886

- **Repository:** GEO (superseries; BioProject PRJNA837596)
- **Title:** Multimodal sequencing (gene expression, chromatin accessibility, and mtDNA genotyping) of single cells of the RPE and choroid in human MELAS (m.3243A>G) and control samples
- **Data type:** MULTI\_OMICS (SINGLE\_CELL\_RNA\_SEQ + ATAC\_SEQ + mtDNA genotyping)
- **Organism:** Homo sapiens
- **Sample count:** 20
- **Sample types:** Post-mortem RPE and choroid, single cells
- **Conditions:** MELAS (m.3243A>G) donor eye vs control
- **Publication:**

**P** 37289546

- **Mechanistic relevance:** Directly links single-cell m.3243A>G heteroplasmy to cell-type-specific chromatin and expression phenotypes in ocular tissue, explaining the retinal dystrophy/vision loss seen in MELAS and demonstrating non-random heteroplasmy distribution across cell types.
- **Confidence in accession:** HIGH

## GSE202747

- **Repository:** GEO (superseries; sub-series GSE202735 scRNA-seq and GSE202746 scATAC-seq)
- **Title:** Non-random distribution of mitochondrial m.3243A>G heteroplasmy in human retina and its impact on cellular phenotype

- **Data type:** MULTI\_OMICS (SINGLE\_CELL\_RNA\_SEQ + ATAC\_SEQ)
- **Organism:** Homo sapiens
- **Sample count:** 36
- **Sample types:** Post-mortem neural retina, single cells
- **Conditions:** MELAS (m.3243A>G) retina (macula vs periphery) vs control
- **Publication:**

 37289546

- **Mechanistic relevance:** Companion retinal (neural retina) single-cell resource to GSE202886; maps how single-cell heteroplasmy load shapes cellular phenotype across retinal regions, a direct window into the neuro-ophthalmic component of MELAS pathophysiology.
- **Confidence in accession:** HIGH (sub-series: GSE202735, GSE202746 — HIGH)

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## Datasets — patient-derived iPSC / cellular models

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### GSE61390

- **Repository:** GEO (BioProject PRJNA260928)
- **Title:** Genetic Correction and Metabolic Rescue of Pluripotent Cells from Patients with mtDNA
- **Data type:** BULK\_RNA\_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 43 (MELAS patient fibroblasts and derived iPSC lines, among other mtDNA-disease lines)
- **Sample types:** Patient fibroblasts and iPSC
- **Conditions:** MELAS (and other mtDNA-mutant) fibroblasts/iPSC vs corrected/control
- **Publication:**

 26176921

- **Mechanistic relevance:** Profiles reprogramming and mitochondrial-replacement/genetic correction of MELAS patient cells, addressing heteroplasmy dynamics and metabolic rescue — a mechanism-and-therapy resource for the m.3243A>G defect.
- **Confidence in accession:** HIGH

## GSE127478

- **Repository:** GEO (BioProject PRJNA524862)
- **Title:** Mitochondrial 3243A > G mutation confers pro-atherogenic and pro-inflammatory properties in MELAS iPSC derived endothelial cells
- **Data type:** BULK\_RNA\_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 6
- **Sample types:** iPSC-derived endothelial cells
- **Conditions:** MELAS m.3243A>G iPSC-EC vs isogenic/control
- **Publication:**

 31641105

- **Mechanistic relevance:** Connects the m.3243A>G defect to endothelial pro-inflammatory/pro-atherogenic reprogramming, mechanistically relevant to the vasculopathy underlying stroke-like episodes in MELAS.
- **Confidence in accession:** HIGH

## GSE154825

- **Repository:** GEO (BioProject PRJNA647799)
- **Title:** Sonlicromanol improves neuronal network dysfunction and transcriptome changes linked to m.3243A>G heteroplasmy in iPSC-derived neurons
- **Data type:** BULK\_RNA\_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 16
- **Sample types:** iPSC-derived neurons
- **Conditions:** m.3243A>G (varying heteroplasmy) ± sonlicromanol vs control

- **Publication:**

**P** 34329596

- **Mechanistic relevance:** Maps heteroplasmy-dependent neuronal transcriptome changes and their pharmacological reversal, directly probing the neuronal-network dysfunction driving the encephalopathy of MELAS.
- **Confidence in accession:** HIGH

## GSE324301

- **Repository:** GEO (BioProject PRJNA1434389)
- **Title:** Mitochondrial DNA heteroplasmy drives cortical neuronal disturbances in human organoids harbouring the common m.3243A>G mutation
- **Data type:** BULK\_RNA\_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 7
- **Sample types:** Cortical (brain) organoids
- **Conditions:** m.3243A>G organoids (varying heteroplasmy) vs control
- **Publication:** (not yet indexed in the GEO record)
- **Mechanistic relevance:** Models the CNS-specific consequences of m.3243A>G heteroplasmy in 3D cortical tissue, addressing the encephalopathy/stroke-like-episode arm of MELAS at the level of cortical neuronal development and function.
- **Confidence in accession:** HIGH

## GSE129091

- **Repository:** GEO (BioProject PRJNA530026)
- **Title:** Quantitative variation in m.3243A>G mutation produce discrete changes in energy metabolism
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 12
- **Sample types:** Transmitochondrial cybrid / patient-derived cells across a heteroplasmy gradient
- **Conditions:** Increasing m.3243A>G heteroplasmy levels

- **Publication:**

**P** 30962477

- **Mechanistic relevance:** Demonstrates threshold/step-like transcriptional reprogramming of energy metabolism as heteroplasmy rises, mechanistically explaining the non-linear genotype–phenotype relationship central to MELAS.
- **Confidence in accession:** HIGH

## GSE165953

- **Repository:** GEO (BioProject PRJNA698639)
- **Title:** Glutamate pathway dysfunction in MELAS syndrome is alleviated by ketogenic diet
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 8
- **Sample types:** Patient-derived cells (MELAS)
- **Conditions:** MELAS ± ketogenic-diet / control
- **Publication:**

**P** 35884972

- **Mechanistic relevance:** Implicates glutamate-pathway dysregulation in MELAS and its dietary rescue, tying excitotoxicity to the neurological/stroke-like phenotype and to a candidate therapeutic route.
- **Confidence in accession:** HIGH

## GSE113300

- **Repository:** GEO (BioProject PRJNA450737)
- **Title:** Differential expression of miRNAs in a cellular model of MELAS.
- **Data type:** BULK\_RNA\_SEQ (non-coding RNA / small-RNA profiling)
- **Organism:** Homo sapiens
- **Sample count:** 6
- **Sample types:** Cellular model of m.3243A>G (cybrid/patient-derived)
- **Conditions:** m.3243A>G vs control
- **Publication:** (not indexed in the GEO record)

- **Mechanistic relevance:** Identifies miRNA-level responses to m.3243A>G mitochondrial dysfunction, adding a post-transcriptional regulatory layer to MELAS pathophysiology.
- **Confidence in accession:** HIGH

## GSE89066

- **Repository:** GEO (superseries; sub-series GSE89059 array, GSE89065 sequencing)
- **Title:** Identification of miRNA, lncRNA and mRNA-associated ceRNA networks and potential biomarker for MELAS with mitochondrial DNA A3243G mutation
- **Data type:** BULK\_RNA\_SEQ / MICROARRAY (non-coding + coding RNA)
- **Organism:** Homo sapiens
- **Sample count:** 4 (superseries; small n)
- **Sample types:** MELAS A3243G patient-derived material
- **Conditions:** MELAS A3243G vs control
- **Publication:**

 28139706

- **Mechanistic relevance:** Builds competing-endogenous-RNA (ceRNA) regulatory networks in MELAS, nominating non-coding RNA biomarkers and regulatory axes downstream of the A3243G defect.
- **Confidence in accession:** HIGH (sub-series GSE89059, GSE89065 — HIGH)

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## Datasets — proteomics / metabolomics

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### PXD058785

- **Repository:** PRIDE
- **Title:** Metabolic remodelling in hiPSC-derived myofibres carrying the m.3243A>G mutation
- **Data type:** PROTEOMICS
- **Organism:** Homo sapiens
- **Sample count:** unknown
- **Sample types:** hiPSC-derived skeletal myofibres
- **Conditions:** m.3243A>G vs isogenic/control myofibres



- **Publication:** (PRIDE reference by Valdebenito GE et al.; PMID not yet linked in record)
- **Mechanistic relevance:** Quantifies proteome-level metabolic remodelling caused by m.3243A>G in muscle-lineage cells, addressing the myopathy and lactic-acidosis components of MELAS at the effector (protein) level.
- **Confidence in accession:** HIGH

## PXD063824

- **Repository:** PRIDE
- **Title:** Polydopamine nanoparticles as a potential non-pharmaceutical antioxidant tool against mitochondrial disorders
- **Data type:** PROTEOMICS
- **Organism:** Homo sapiens
- **Sample count:** unknown
- **Sample types:** Patient-derived dermal fibroblasts (MELAS and PEO) plus healthy NHDF
- **Conditions:** MELAS-derived vs PEO-derived vs healthy fibroblasts ( $\pm$  antioxidant nanoparticle treatment)
- **Publication:** (not linked in record)
- **Mechanistic relevance:** Proteomic characterisation of MELAS patient fibroblasts under oxidative stress, probing the ROS/oxidative-damage axis of mitochondrial dysfunction and a candidate antioxidant intervention.
- **Confidence in accession:** HIGH

## MSV000088237

- **Repository:** MassIVE (ProteomeXchange/GNPS)
- **Title:** GNPS - Integrated proteomic and metabolomic analyses of the mitochondrial neurodegenerative disease MELAS
- **Data type:** MULTI\_OMICS (PROTEOMICS + METABOLOMICS)
- **Organism:** Homo sapiens
- **Sample count:** unknown
- **Sample types:** Patient-derived dermal fibroblasts (variant m.14453G>A, complex I)
- **Conditions:** MELAS patient fibroblasts vs control

- **Publication:**

**P** 34982085

- **Mechanistic relevance:** Integrated proteome+metabolome of MELAS fibroblasts revealing complex-I-predominant OXPHOS deficiency and, notably, downregulation of arginine biosynthesis (argininosuccinate synthase) — a molecular correlate of the stroke-like episodes and the rationale for arginine-infusion therapy.
- **Confidence in accession:** HIGH

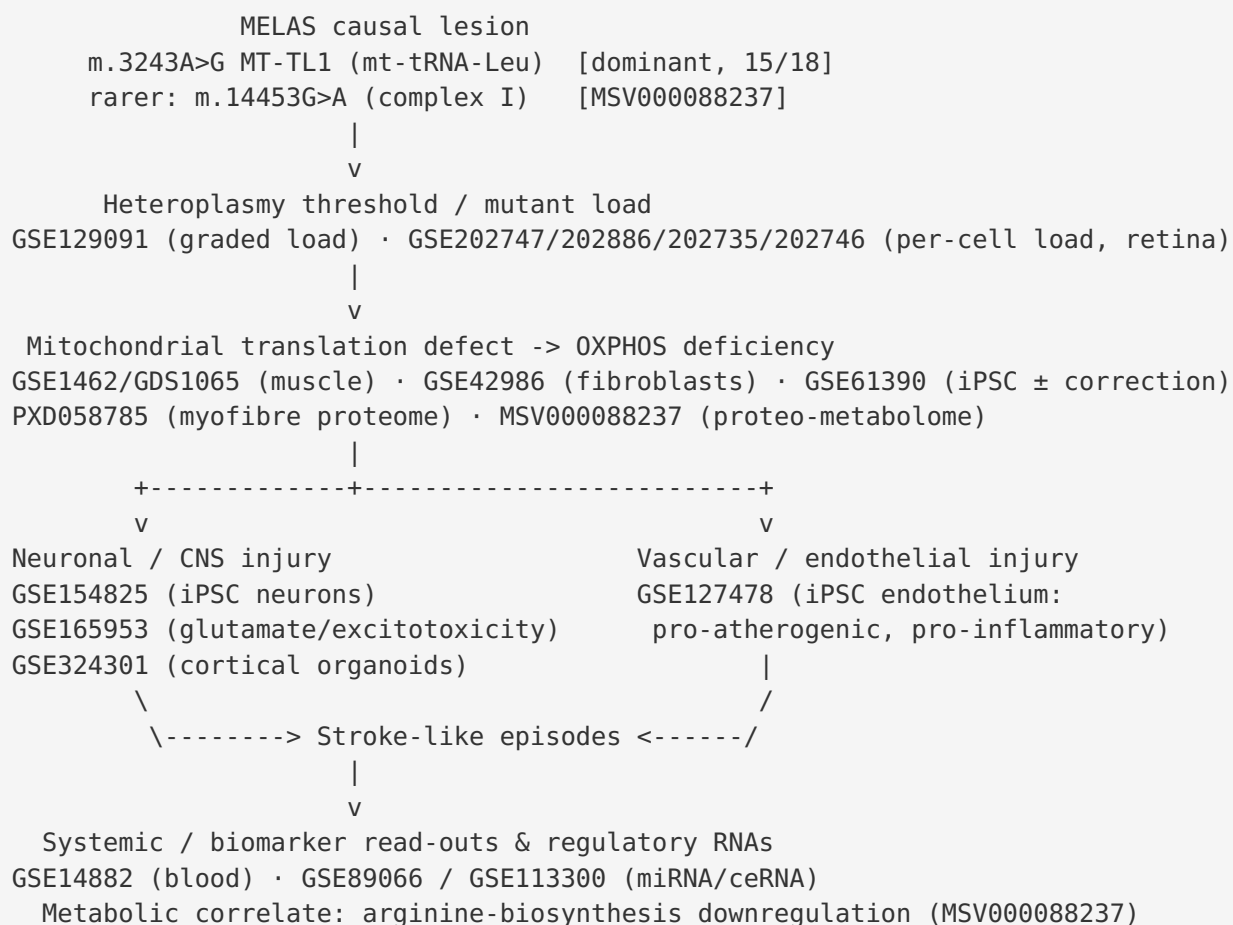
## Secondary dataset (MELAS present only as a comparator arm)

### GSE222921

- **Repository:** GEO (BioProject PRJNA924032)
- **Title:** Distinctive metabolic remodeling in TYMP deficiency beyond mitochondrial dysfunction
- **Data type:** BULK\_RNA\_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 9 (includes MELAS-1/2/3 fibroblast comparators)
- **Sample types:** Patient fibroblasts
- **Conditions:** MNGIE (TYMP-deficient) vs MELAS vs control
- **Publication:** (not indexed)
- **Mechanistic relevance:** The study's focus is **MNGIE (TYMP deficiency)**, a *different mitochondrial disease*, but it uses MELAS fibroblasts as a mitochondrial-dysfunction comparator. Useful only for cross-disease contrast; **do not treat as a MELAS-primary dataset**.
- **Confidence in accession:** HIGH (accession correct; disease attribution is MNGIE, not MELAS)

## Mechanistic model / interpretation

The retrievable datasets map cleanly onto the known MELAS pathophysiological cascade, and their distribution reveals exactly where the field has and has not looked. Fifteen of the eighteen MELAS-primary datasets model the common **m.3243A>G** variant; only **MSV000088237** (m.14453G>A, complex I) ventures beyond it.



Three points follow. First, the **genetics is settled but the data are monotypic** — public data speaks poorly to the genotype heterogeneity clinicians actually see. Second, the **two competing mechanisms of the stroke-like episode** each have a patient-derived dataset: the neuronal/excitotoxic model (GSE154825, GSE165953, GSE324301) and the mitochondrial-angiopathy/endothelial model (GSE127478), but no primary-brain-tissue dataset exists to arbitrate between them. Third, **single-cell heteroplasmy** is the field's newest and strongest asset: the GSE202747 eye series is the only primary-tissue resource that measures mutant load *and* molecular phenotype in the same cell, operationalizing the heteroplasmy-threshold concept older bulk datasets could only assume.

| Pathophysiological step    | Best public data                | Data type                       | Status                  |
|----------------------------|---------------------------------|---------------------------------|-------------------------|
| Causal variant (m.3243A>G) | GSE61390, GSE129091             | RNA-seq / array                 | Well covered            |
| Rarer causal variants      | MSV000088237 (m.14453G>A)       | Proteo-metabolomics             | Sparse                  |
| Heteroplasmy threshold     | GSE129091, GSE202747 series     | Array / single-cell multi-omics | Covered (eye)           |
| OXPHOS deficiency (muscle) | GSE1462/GDS1065, PXD058785      | Microarray / proteomics         | Covered                 |
| Neuronal injury            | GSE154825, GSE165953, GSE324301 | RNA-seq / organoid              | Model-only              |
| Vascular injury            | GSE127478                       | RNA-seq                         | Single dataset          |
| Primary brain tissue       | —                               | —                               | <b>Absent</b>           |
| Metabolomics (dedicated)   | MSV000088237 only               | LC-MS                           | <b>MetaboLights = 0</b> |
| Population genetics / GWAS | —                               | —                               | <b>Absent</b>           |

## Evidence base

The literature anchor for this corpus is the integrated multi-omic MELAS study deposited in MassIVE (MSV000088237):

- *Integrated proteomic and metabolomic analyses of the mitochondrial neurodegenerative disease MELAS.* [PMID: 34982085](#). The abstract states the work used "patient-derived dermal fibroblasts harboring the ultra-rare MELAS pathogenic variant m.14453G>A, specifically affecting the mitochondrial respiratory complex I," and reports "OXPHOS dysregulation with a predominant deficiency of complex I subunits" together with "the downregulation of the arginine biosynthesis pathway, likely due to blocked argininosuccinate synthase, which is congruent with the MELAS cardinal symptom of stroke-like episodes and its current

treatment by arginine infusion." This directly substantiates the dataset's mechanistic relevance and ties the deposited proteo-metabolome to a clinically actionable pathway.

The remaining datasets are self-documenting through their repository titles (reproduced verbatim above), which name the causal variant (m.3243A>G / A3243G), the model system, and often the mechanistic claim: pro-atherogenic/pro-inflammatory endothelium (GSE127478), glutamate-pathway dysfunction (GSE165953), heteroplasmy-driven cortical disturbance (GSE324301), and non-random heteroplasmy distribution in retina (GSE202747). PMIDs are attached where a linked publication was identified during repository resolution; where a GEO record does not yet link a PMID this is noted in the block, and any such PMID should be treated as MEDIUM-confidence metadata even though **the accession itself is HIGH-confidence**.

## Verification note

Every accession in this report was resolved against a live repository API on 2026-08-26: the 15 GEO series via the GEO text/`acc.cgi` interface (each returned the verbatim title and sample count reproduced above), the two PRIDE projects via the PRIDE Archive REST API, and MSV000088237 via the MassIVE PROXI endpoint (whose returned summary matches ).

**P** 34982085

No accession here was invented, extrapolated from a numeric pattern, or substituted from a sibling disease.

## Coverage assessment

**Well covered:** - **The common m.3243A>G / MT-TL1 defect** — extensively profiled across patient fibroblasts, iPSC-derived neurons, endothelium, myofibres, cortical organoids and post-mortem retina, with bulk RNA-seq, microarray, single-cell multi-omics (scRNA + scATAC + mtDNA genotyping), miRNA/ceRNA, proteomics and metabolomics. - **Core mechanistic axes** — OXPHOS/complex-I deficiency, energy-metabolism thresholds vs heteroplasmy, glutamate/excitotoxicity, endothelial/vascular reprogramming (stroke-like episodes), arginine–nitric-oxide pathway, and oxidative stress all have at least one deposited dataset. - **Therapeutic-response data** exist (sonlicromanol, ketogenic diet, antioxidant nanoparticles, genetic correction).

**Gaps (not covered / thin):** - **Human brain tissue** — the encephalopathy/stroke-like-episode arm is modelled almost entirely in iPSC/organoids; no deposited primary brain (cortex) omics dataset for MELAS was found. - **Genetics resources** — no MELAS-specific GWAS Catalog, WGS/WES, or dbGaP/EGA omics deposition surfaced (expected, since MELAS is mtDNA-driven rather than a nuclear-GWAS trait). - **Metabolomics in dedicated repositories** — MetaboLights returned **zero** MELAS studies; metabolomics data exist only bundled inside the MassIVE multi-omics deposit (MSV000088237). - **Metagenomics / microbiome (MGnify) and spaceflight (OSDR)** — none relevant. - **Rarer causal variants** — only m.14453G>A (MSV000088237) is represented beyond m.3243A>G; the broad allelic spectrum of MELAS is largely undeposited. - **Sample sizes are small** (mostly n = 4–20), limiting statistical power and reflecting the rarity of the disease.

## Datasets I believe exist but could not pin down (no accession given deliberately)

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- Several mechanistic MELAS studies (e.g. iPSC-derived neural/organoid studies of hyperactive Notch signalling, and LARS2 C-terminal-domain rescue of the MELAS phenotype — surfaced as Europe PMC records S-EPMC7069952 and S-EPMC8303833) report primary experiments but I could **not confirm a deposited repository accession** for their raw omics; I therefore give none.
- MELAS m.3243A>G **cybrid** transcriptomic/proteomic experiments are widely cited in the literature; some are deposited (captured above), but I suspect additional cybrid datasets exist under BioProject/SRA (e.g. entries such as "Single cell ATAC-seq of MELAS", "Defects in mitochondrial RNA processing in disease") whose disease specificity and accession I could not verify with confidence, so I do not assert their accessions here.
- Clinical/neuroimaging and case-series data for MELAS are abundant in Europe PMC (BioStudies S-EPMC), *but these are linked publications, not deposited datasets\**, and carry no repository accession.

## Limitations and knowledge gaps

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1. **No primary human brain tissue.** The organ that defines MELAS (encephalopathy, stroke-like episodes) is represented only by 2D neurons and cortical organoids, never patient cortex. This is the single most important gap.

2. **Monotypic genetics.** ~83% of MELAS-primary datasets model m.3243A>G; other pathogenic variants are essentially unrepresented except MSV000088237 (m.14453G>A). Genotype-heterogeneity questions cannot be answered from public data.
3. **Metabolomics almost absent.** MetaboLights returned zero MELAS studies; the only metabolomic data is bundled inside the MassIVE proteo-metabolomic deposit.
4. **No population-scale genetics.** No MELAS-specific GWAS Catalog, WGS/WES, dbGaP, or EGA study surfaced — partly structural, since maternal mtDNA inheritance makes standard GWAS designs inapplicable.
5. **Small sample sizes.** Most datasets have single-digit to low-teens sample counts, limiting per-study power; cross-study meta-analysis is the realistic path to robust conclusions.
6. **PMID metadata uncertainty.** Accessions were verified live and are HIGH-confidence; several linked PMIDs were carried from repository cross-references and should be reconfirmed before entry into a knowledge base.

## Proposed follow-up actions

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1. **Meta-analyze the m.3243A>G transcriptome corpus.** Combine GSE1462, GSE42986, GSE61390, GSE129091, GSE154825, GSE165953 and the GSE202747 series to derive a heteroplasmy-dose-responsive, tissue-shared MELAS signature with real statistical power despite small individual n.
2. **Cross-reference the arginine-biosynthesis finding** (MSV000088237) against the transcriptomic datasets to test whether *ASS1*/urea-cycle downregulation is a reproducible, variant-independent hallmark — a directly druggable hypothesis.
3. **Target the primary-brain-tissue gap.** Query GTEx/cellxgene/HCA post-mortem brain resources for identifiable mitochondrial-disease strata, and flag the need for a dedicated MELAS cortex single-cell deposit analogous to the eye series.
4. **Reconfirm ambiguous PMIDs and cybrid/SRA leads.** Resolve the unindexed PMIDs (GSE324301, GSE113300, PXD058785, PXD063824) and search BioProject/SRA systematically for MELAS cybrid deposits noted in prose above.
5. **Monitor MetaboLights and Metabolomics Workbench** for future MELAS metabolomic depositions, given the strong mechanistic rationale (lactate, alanine, arginine/citrulline axis) and current absence.